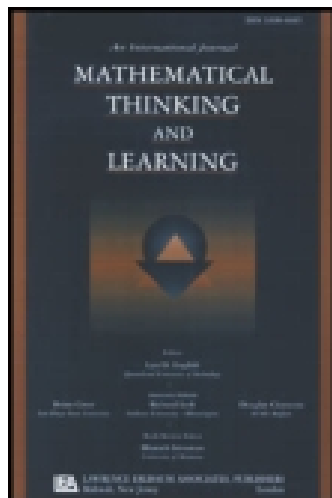


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How We Think: A Theory of Goal-Oriented Decision Making and Its Educational Applications, by Alan H. Schoenfeld

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BOOK REVIEW

Alan H. Schoenfeld (2011). *How We Think: A Theory of Goal-Oriented Decision Making and Its Educational Applications*. New York, NY: Routledge. ISBN: 978-0-415-87865-4. xviii+245 pp., \$46.95.

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Classrooms have the appearance of orchestrated chaos. Teachers make instructional moves and students reply; students initiate some action and teachers respond. Experienced educators understand that there is a rhythm, a tempo, evident in all that instructional chaos. However, it is not a rhythm that can often be precisely or reliably reproduced, or even readily anticipated. Certainly there are routines or subroutines, what Schoenfeld and others refer to as scripts or schemas (Anderson, Reynolds, Schallert, & Goetz, 1977; Mandler, 1984; Mason & Morris, 2010), that can be recognized and named. For instance, there is the prologue, where teachers introduce the lesson to follow and students acknowledge their expected role in that lesson. There is the review or wrap-up, when students are called on to communicate what they have already learned or can perform or the teacher reiterates what was “learned” in the preceding activities. But overall, the flow of the entire lesson cycle far overruns these particular recognizable instructional episodes and far outreaches one’s abilities to predict what teachers will do next or why such action will be taken. Into this apparent orchestrated chaos steps Alan H. Schoenfeld with his proposed theory of Goal-Oriented Decision-Making.

THE GENERAL STRUCTURE OF *HOW WE THINK*

As Schoenfeld seeks to establish in his volume, *How We Think: A Theory of Goal-Oriented Decision Making and its Educational Applications*, there is more predictability than meets the eye in the instructional flow of the classroom. His contention is that if we, as perceptive observers, enter the classroom chaos armed with critical information about a specific teacher, then the instructional movements that teacher makes and the choices those movements expose become more predictable and more understandable—less chaotic in appearance. Specifically, building on decades of his empirical work on problem solving, especially in the domain of mathematics, Schoenfeld sets out to make a strong case that teachers’ instructional decision making—a form of problem solving—is a function of the goals they set for the instruction, the resources upon

which they can draw, and the orientations they carry into the classroom and to that particular lesson. The instructional goals or purposes can be conscious and intended or they may be more tacit in form. The resources to which Schoenfeld refers encompass primarily the teachers' knowledge about the domain, task, and pedagogy. Moreover, the orientations component he identifies encompasses what teachers hold to be true about the domain, teaching, and the learner.

To use Schoenfeld's own words, the claim "at the heart" of this book is the following:

If you can understand (a) the teacher's agenda and the routine ways in which the teacher tries to meet the goals that are implicit or explicit in that agenda, and (b) the factors that shape the teacher's prioritizing and goal setting when potentially consequential unforeseen events arise, then you can explain how and why teachers make the moment-by-moment choices they make as they teach. (p. 10)

Throughout the chapters of *How We Think*, Schoenfeld delves into each of these key areas. He thoughtfully explains each (Chapters 1–3) and describes how they form the bases of his theory of goal-oriented decision making. Next, he illustrates the influence of goals, resources, and orientations as well as the ecological validity of his theoretical model through detailed analyses of transcripts of mathematics and science lessons (Chapters 4–6). For instance, in Chapter 6, he meticulously walks us through a segment of a lesson in a third-grade classroom dealing with the properties of even and odd numbers. The lesson he analyzes was taught by Deborah Ball, a renowned mathematics educator and researcher. He uses the transcripts from this lesson segment to illustrate how what he initially judged as surprising pedagogical movements by Ball became increasingly clearer once he began to take into account her goals, resources, and her espoused orientations.

In the closing chapters of the volume (Chapters 7 and 8), Schoenfeld moves away from the territories of mathematics and science and ventures into the field of medical consultation. His intentions appear to demonstrate the broad applicability of his theory to domains that appear appreciably different from mathematics or science education. He also wanted to consider joint problem solving that arises between doctor and patient during the consultation process. The intriguing feature of this extension and generalization is that Schoenfeld delves into a personal medical issue, adult onset (Type II) diabetes, and his interactions with a doctor with whom he is very familiar.

So, does Schoenfeld succeed in his laudable intentions? Does he craft a convincing and compelling argument for his theory of goal-oriented decision making? To answer that question, I want to loosely apply Schoenfeld's three key dimensions, goals, resources, and orientations, not to ascertain whether I understand what he did or why—since he thoughtfully details all of that for the reader—but rather to judge whether his goals and associated decisions worked for me as a reader of this volume. Specifically, I ask how effectively Schoenfeld's goals and orientation are manifest within the pages of this volume; whether the treatment of resources, especially with regard to domain and pedagogical knowledge, is sufficiently represented; and how well the process of decision-making is explicated and justified.

ORIENTATION EVIDENT AND GOALS REALIZED?

If I were to characterize the orientation represented within the pages of *How We Think*, I would have to borrow a term I have used to signify my own orientation to learning, cognitive

contextualism (Alexander, Schallert, & Reynolds, 2009); an orientation that appreciates the thinking and reasoning of the individual mind while duly acknowledging the power of the social collective and contextual factors. One of the evident strengths of this book is its ability to speak to the diverse community of mathematics and science education as a consequence of this multidimensional orientation to human learning and performance. For instance, the meticulous moment-by-moment analysis of skilled teachers' pedagogical movements in light of their goals, orientation, and knowledge is evidence of Schoenfeld's extensive understanding of human cognition and his roots in both mathematics and cognitive science. Few could have brought the requisite analytic structure and perspective to these mathematics and physics lessons.

It is also apparent that Schoenfeld's orientation toward decision making is not restricted to the mental ruminations of a teacher operating in a sociocontextual vacuum. Each step in the decision-making chain is shaped by the actions of others and by the co-constructed understandings that arise. His theory takes into account contingencies of the classroom environment. Clearly, his theory of goal-oriented decision making would not be meaningful without this sociocontextual dynamic.

Of course, orientation, for Schoenfeld, is not solely a matter of one's beliefs about the nature of learning in general, but also about mathematics or science and how those subjects should be taught. For instance, considering a lesson segment on blood alcohol content, Schoenfeld encapsulates the orientation of Jim Minstrell, a recipient of the Presidential Award for Excellence in Mathematics and Science Teaching, thus:

Minstrell conceives of physics (and thus of physics learning) as a sense-making activity, and he believes that students' classroom experiences should support the development of their abilities as sense makers. . . . Minstrell believes that his students need to engage directly with the material and not simply be told what works. Hence Minstrell works to create a discourse community that involves small-group work among students, and he structures teacher-student exchanges in a way that minimizes "telling." (pp. 90–91)

Overall, I find that Schoenfeld's perspective and the evidence that he offers in its support work best when the potential audience is educational researchers interested in identifying the patterns within teachers' instructional actions. I am not convinced of the practical utility of this theoretical model for practicing teachers seeking to understand their own decision making or that of their peers. Decisions within the classroom are simply too numerous and multifaceted in their nature, and progress at such a rate and tempo that only a few lesson segments could be systematically dissected in such a pedagogical postmortem as we find in these chapters. Likewise, while I applaud Schoenfeld's attempt to move the theory away from mathematics and physics teaching and beyond the confines of the classroom, and even though the interactions between doctor and patient are both thoughtfully and sensibly analyzed, I see this effort to generalize the model as less convincing or compelling. The book and the theoretical model are at their best when the focus remains squarely on the thinking and reasoning of mathematics or science teachers during the course of instructional decision making.

The aforementioned criticisms are not meant to cast doubts on the contributions of this theoretical work in general or this volume in particular. Mathematical researchers can benefit greatly from understanding the parameters of Schoenfeld's theory and the relations his theoretical model hypothesizes, as well as from the data-analytic system he has applied in capturing classroom interactions. Indeed, the method of transcribing, representing, and interpreting classroom discourse

illustrated within the chapters of *How We Think* stands as a model for those interested in capturing real-time sociocontextual dynamics in reliable and informative ways.

KNOWLEDGE AND ASSOCIATED RESOURCES REPRESENTED?

Two aspects of Schoenfeld's presentation of resources in *How We Think* merit discussion here. First, while Schoenfeld uses the more general word, resources, to characterize a significant aspect of goal-directed decision making, the discussion throughout the book and in the research upon which this book is based deals almost exclusively with knowledge. Given Schoenfeld's impressive body of work on mathematical problem solving within the educational psychological literature, it was not surprising to see domain and pedagogical knowledge so prominently displayed within the resource dimension of his goal-oriented decision-making theory. Further, it is hard to take issue with any of the knowledge principles outlined in the introduction of this volume and that frame subsequent analyses in the ensuing chapters. Insights into the knowledge base of teachers can go a long way to explain what they perceive within the flow of instruction and to what they allocate their attention.

The discussion of Mark Nelson's Algebra I lesson in Chapter 4 solidly positions the influence of teacher domain and pedagogical knowledge in decision making. In this chapter, Schoenfeld expertly deconstructs this student teacher's hour-long session on dividing expressions with exponents and makes the following observations regarding Nelson's resources.

The arithmetic of exponents was familiar territory for Nelson, who was well prepared mathematically. The pedagogical resource he relied on in give-and-take sessions with students was a questioning strategy in which he asked students for the answers to problems he had given them and then provided feedback on their responses. . . . Finally, given that this was the first time teaching this particular material, Nelson could not be expected to have a reservoir of pedagogical content knowledge for this topic. . . . Such knowledge is developed with experience, if it is developed at all. (p. 77)

A second aspect that warrants discussion pertains to the orientation toward knowledge that the goal-oriented decision-making theory represents. One of the clear strengths of this book is the treatment of content and pedagogical knowledge at the declarative, conceptual, and procedural levels. With each detailed analysis, Schoenfeld provides persuasive evidence of how expertise and its associated knowledge come into play in the decision making of the teachers—Nelson, Minstrell, and Ball—and Dr. B. The relations between expertise and decision making have been well documented by others within the educational literature (Alexander, 2003; Ericsson, Charness, Feltovich, & Hoffman, 2006). But, few have illustrated that relation with the precision and care found in the pages of this book.

That being said, although Schoenfeld's treatment of orientation within the chapters of this book pays homage to individuals' beliefs about the nature of their domains, there is much theoretical and empirical work on epistemology or epistemic beliefs that could add to this dimension of the overall theoretical model. There are certainly implied references to teachers' and students' beliefs about knowledge and knowing woven throughout *How We Think*, but no explicit discussion of such beliefs or to the research dealing with epistemic beliefs in the domains of

mathematics or science (e.g., diSessa, Elby, & Hammer, 2002; Schommer-Aikins, Duell, & Hutter, 2005).

DECISION-MAKING EXPLICATED AND JUSTIFIED?

What is essential to keep in mind is that the theoretical model Schoenfeld proffers pertains to goal-oriented decision making in well-practiced, knowledge-intensive domains. What cannot be explicated by this theory or by any other cognitive or sociocognitive model are the spontaneous choices that individuals repeatedly make whether in classrooms, in doctor's offices, or with any tasks that are not highly proceduralized or routinized. Many human actions, including decisions, do not arise from conscious or rational thought. If there is rationalization involved in such choices, it may come in the form of post hoc justifications or explanations. Goal-oriented decision making rests on the assumption that decisions on the part of the teacher are the result of a deliberate and conscious process. Some portion of those goal-directed decisions fall in the class that Schoenfeld classifies as routine decision making, "relatively unproblematic" forms of decision making that are instantiations of well-practiced scripts and schemes. The challenge, as Schoenfeld rightly notes, is when the routine scripts or schemes cannot be fully enacted, as when nonroutine situations are encountered.

Through all the step-by-step analyses of transcripts from beginning or master teachers engaged in mathematics or physics instruction or from an established physician involved in doctor-patient consultation, accompanied by numerous graphics and flow charts, Schoenfeld methodically builds his case for his theory. Interestingly, as with others in the educational community seeking to problematize mathematics and science instruction, Schoenfeld is determined to reduce the noise within classroom data—to find a workable path through the labyrinth.

TAKING STOCK, APPLICATIONS, AND NEXT STEPS

Taking Stock

So the lingering question remains whether the goals Schoenfeld set for himself and for this volume were achieved. Do I find this multidimensional orientation toward decision making viable? Am I persuaded by the detailed cases he offers and is his proof compelling and irrefutable? And most importantly for this review, do I regard this compilation and extension of Schoenfeld's years of research to be a work that belongs in the library of educators concerned with teaching and learning?

I concur with Schoenfeld when he states that, "theories and models in science are always subject to revision and refinement," that they must be built on "certain assumptions about the nature of mental organization," (p. 58), and that "all models are, in a serious sense, metaphors" (p. 60). Moreover, even when one attempts to constrain the problem space to well-practiced, highly procedural tasks undertaken within knowledge-intensive domains, one cannot hope to address all factors at play either within the mind of the individual or within the sociocultural context. Nonetheless, Schoenfeld does succeed in discerning patterns within teacher-student

and patient-client exchanges that are certainly informed by the orientations and goals of the decision-maker and reflective of the knowledge that they bring to the task and the context.

Applications

I am aware that the theory that Schoenfeld presents in *How We Think* has been used as the basis for professional development in order to assist teachers in problematizing their own practice (e.g., Arcavi & Schoenfeld, 2008). Even so, I contend that the knowledge and expertise required to apply the goal-oriented decision making model render the theory more suitable for researchers in mathematics and science and having more appealing for those with some form of cognitive orientation to human learning and development than for practicing teachers. For teachers faced with the ongoing demands of classroom life, the model best serves as a general tool for self-reflection—a catalyst for considering what they elect to do and why those decisions may have been made. In effect, the application for practitioners is as a critical reminder to stop, put their routine decision making under the microscope, and scrutinize their actions with greater perception and awareness than might otherwise be brought to bear.

Next Steps

Understanding the moment-by-moment decisions that teachers make when engaged in well-practiced knowledge-intensive lessons remains an invaluable contribution to the education literature. However, even a complete theory of how this unfolds within the dynamic context only touches the surface of what occurs in classrooms—even mathematics or physics classrooms like those represented in this volume. Although Schoenfeld did venture outside the classroom in order to demonstrate the viability of his theory when the task is a doctor-patient consultation, this extension raised more questions for me than it answered.

For instance, what the doctor-patient consultation illustrated is that a model of decision making become less containable within any interpersonal exchanges. Specifically, when decision making occurs between multiple players, whether teacher and students or doctor and patient, then each comes to the interaction with their own set of goals, orientations, resources, and decision-making processes. Control of the decision cannot rest in the hands of any one actor or be understood as such. I look forward to more consideration of the duality or multiplicity in decision making and consider Schoenfeld's goal-oriented decision making theory an excellent starting point for this pursuit.

FINAL THOUGHTS

As someone who has long admired the insights that Alan Schoenfeld has brought to the educational research community, especially with regard to mathematical problem solving and the interplay of conceptual and procedural knowledge in academic development, I see *How We Think* as a tribute to Schoenfeld and a natural extension of decades of his theoretical and empirical work. I learned much from reading its chapters and find many aspects of the goal-oriented decision-making theory compelling. Without question, there are many in the educational research

community who could likewise gain from the wisdom shared in its pages. Thus, for anyone invested in understanding how pedagogical decision making happens in the course of teaching and learning and why, or for those who are seeking a viable methodology for delving into the moment-by-moment decisions of teachers, *How We Think* is a must-have scholarly volume.

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